Practice workbook · Sum to infinity of a geometric series

Using S = a/(1 - r) once a and r are known or found, sometimes in surd or fraction form.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-13

1. Question 8 · part c · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c only.

8 The sum of the first and second terms of a geometric series G is 400

The sum of the second and third terms of G is 100

(a) Show that the common ratio of G is $\frac{1}{4}$

(4)

(b) Show that the first term of G is 320

(2)

(c) Find the sum to infinity of G

(2)

The sum to n terms of G is S_n

(d) Find, using logarithms, the least value of n such that

$$S_n > 426.6$$

(4)

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1. Question 8 · part c · 2 marks · Working and answer

Working and answer

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2. Question 3 · part b · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

3 A geometric series G has first term a and common ratio r

The 2nd term of G is $\frac{5}{16}$ and the 5th term of G is $\frac{135}{1024}$

(a) Find

(i) the value of r

(ii) the value of a

(5)

Given that G is convergent with sum to infinity S

(b) find the exact value of S

(2)

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2. Question 3 · part b · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 7 marks)

2. Question 3 · part b · 2 marks · Working and answer

Working and answer

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3. Question 5 · part a · 2 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

5 A geometric series G has first term 12 and common ratio $\frac{3}{8}$

(a) Find the sum to infinity of G

(2)

(b) Show that the 6th term of G can be written as $\frac{3^6}{2^{13}}$

(3)

The nth term of G is u_n

(c) By finding an expression for u_n in terms of n, show that

$$\log_2 u_n = n\log_2 3 - 3n + 5$$

(5)

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3. Question 5 · part a · 2 marks · Working and answer

Working and answer

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4. Question 7 · part c · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

7 A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of x.

(5)

Given that $x < 1$,

(b) show that the series is convergent.

(2)

The sum to infinity of the series is S.

(c) Find the value of S.

(2)

The sum of the first n terms of the series is S_n

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of n.

(3)

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4. Question 7 · part c · 2 marks · Working and answer

Working and answer

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5. Question 6 · part b · 3 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

6 The nth term of a geometric series G is U_n

The first three terms of G are given by

$$U_1 = q(4p + 1) \qquad U_2 = q(2p + 3) \qquad U_3 = q(2p - 3)$$

(a) Find the possible values of p

(5)

Given that G is convergent with sum to infinity 250

(b) find the value of q

(3)

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5. Question 6 · part b · 3 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 8 marks)

5. Question 6 · part b · 3 marks · Working and answer

Working and answer

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6. Question 8 · part c · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

8 The sum of the first 2 terms of a geometric series G is 360
The sum of the 2nd and 3rd terms of G is 288

The nth term of G is U_n

(a) Show that $U_n = A\left(\frac{4}{5}\right)^{n-1}$ where A is an integer to be found.

(7)

(b) Explain why G is convergent.

(1)

(c) Hence find the sum to infinity of G

(2)

(d) Find the least number of terms for which the sum is greater than 978

(4)

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6. Question 8 · part c · 2 marks · Working and answer

Working and answer

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7. Question 5 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

5 The nth term of a geometric series with common ratio r is u_n

Given that $u_2 + u_4 = 212.5$ and that $u_3 + u_4 = 62.5$

(a) find the two possible values of r.

(5)

Given that the series is convergent with sum to infinity S,

(b) find the exact value of S.

(2)

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7. Question 5 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.



Question 5 continued

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(Total for Question 5 is 7 marks)

7. Question 5 · part b · 2 marks · Working and answer

Working and answer

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8. Question 7 · part c · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

7 The geometric series G has first term a, common ratio r and nth term u_n

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

(3)

(b) Hence find a in terms of e and x.

(3)

Given that the sum to infinity of G can be written as $\dfrac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for p in terms of x.

(3)

Given that $u_{18} > 1.6$ and that x is an integer,

(d) find the least value of x.

(4)

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8. Question 7 · part c · 3 marks · Working and answer

Working and answer

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9. Question 3 · part b · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

3 A geometric series has first term a and common ratio r, where $r > 0$

Given that the 3rd term of the series is 5 and that the 5th term of the series is $\frac{5}{2}$

(a) find

(i) the exact value of r

(ii) the value of a

(4)

(b) Find the sum to infinity of this series.
Give your answer in the form $p + q\sqrt{2}$ where p and q are integers.

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9. Question 3 · part b · 2 marks

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Further practice · Attempt part b only.



Question 3 continued

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(Total for Question 3 is 6 marks)

9. Question 3 · part b · 2 marks · Working and answer

Working and answer

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10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

7 The 7th term of a geometric series is 192 and the 8th term of this geometric series is 1152

(a) Find, as a fraction in its simplest form, the 4th term of this geometric series.

(3)

A different geometric series G has a common ratio r and nth term t_n

Given that $t_3 = 24$ and $t_2 + t_3 + t_4 = -36$

(b) show that r satisfies the equation

$$2r^2 + 5r + 2 = 0$$

(5)

Given further that G is convergent with sum to infinity S,

(c) find the value of S.

(4)

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10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 12 marks)

10. Question 7 · part c · 4 marks · Working and answer

Working and answer

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11. Question 7 · part b · 2 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

7 A geometric series G with common ratio r, has first term 16 and third term $\frac{2704}{625}$

(a) Find the two possible values of r

(2)

Given that $r > 0$

(b) find the sum to infinity of G

(2)

The sum to n terms of G is greater than 33

(c) Find, using logarithms, the least possible value of n
Show your working clearly.

(5)

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11. Question 7 · part b · 2 marks · Working and answer

Working and answer

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12. Question 2 · part whole · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

2 The nth term of a convergent geometric series is $8^{(1-2n)}$

Find the sum to infinity of the series.

Give your answer in the form $\frac{p}{q}$ where p and q are integers to be found.

(6)

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12. Question 2 · part whole · 6 marks

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Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 6 marks)

12. Question 2 · part whole · 6 marks · Working and answer

Working and answer

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13. Question 4 · part ii · 7 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part ii only.

4 (i) Find the value of $\sum_{k=6}^{17} 3(2)^{k-1}$

(4)

(ii) The common ratio of a geometric series G is positive.

The third term of G is $\frac{7}{13}$ and the ninth term of G is $\frac{448}{9477}$

G is convergent with sum to infinity S

Find the exact value of S

(7)

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13. Question 4 · part ii · 7 marks

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Further practice · Attempt part ii only.

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Question 4 continued

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(Total for Question 4 is 11 marks)

13. Question 4 · part ii · 7 marks · Working and answer

Working and answer

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